

```
In [44]: import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [24, 27, 22, 32],
    'City': ['New York', 'Los Angeles', 'Chicago', 'Houston']
}
df=pd.DataFrame(data)
print(df.head(2))
```

	Name	Age	City
0	Alice	24	New York
1	Bob	27	Los Angeles

```
In [31]: print(df.columns)

Index(['Name', 'Age', 'City'], dtype='object')
```

```
In [32]: print(df.shape)

(4, 3)
```

```
In [33]: print(df.info)

<bound method DataFrame.info of
0    Alice    24    New York
1     Bob    27    Los Angeles
2  Charlie    22     Chicago
3   David    32     Houston>
```

```
In [34]: print(df['Name'])

0    Alice
1     Bob
2  Charlie
3   David
Name: Name, dtype: object
```

```
In [35]: print(df[['Name', 'City']])

      Name      City
0   Alice  New York
1    Bob  Los Angeles
2  Charlie   Chicago
3   David   Houston
```

```
In [36]: print(df.iloc[2])

Name    Charlie
Age        22
City    Chicago
Name: 2, dtype: object
```

```
In [37]: print(df.loc[df['Name'] == 'Charlie'])

      Name  Age  City
2  Charlie   22  Chicago
```

```
In [38]: print(df[df['Age'] > 25])
```

	Name	Age	City
1	Bob	27	Los Angeles
3	David	32	Houston

```
In [39]: print(df[(df['City']=='Chicago') | (df['City']=='Houston')])
```

	Name	Age	City
2	Charlie	22	Chicago
3	David	32	Houston

```
In [40]: print(df[(df['Age'] > 23) & (df['Age'] < 30)])
```

	Name	Age	City
0	Alice	24	New York
1	Bob	27	Los Angeles

```
In [45]: df['Score']=[85,90,88,95]  
print(df)
```

	Name	Age	City	Score
0	Alice	24	New York	85
1	Bob	27	Los Angeles	90
2	Charlie	22	Chicago	88
3	David	32	Houston	95

```
In [46]: df.loc[1,'Age']=28  
print(df)
```

	Name	Age	City	Score
0	Alice	24	New York	85
1	Bob	28	Los Angeles	90
2	Charlie	22	Chicago	88
3	David	32	Houston	95

```
In [48]: df.drop('City',axis=1,inplace=True)  
print(df)
```

	Name	Age	Score
0	Alice	24	85
1	Bob	28	90
2	Charlie	22	88
3	David	32	95

```
In [54]: df.drop(index=3,axis=1,inplace=True)  
print(df)
```

	Name	Age	Score
0	Alice	24	85
1	Bob	28	90
2	Charlie	22	88

```
In [88]: import pandas as pd
data = {
    'Department': ['HR', 'IT', 'HR', 'IT'],
    'Salary': [30000, 50000, 35000, 55000],
    'Experience': [2, 5, 3, 6]
}
df=pd.DataFrame(data)
a=df.groupby('Department')['Salary'].mean()
print(a)
```

```
Department
HR      32500.0
IT      52500.0
Name: Salary, dtype: float64
```

```
In [89]: b=df.groupby('Department')['Experience'].max()
print(b)
```

```
Department
HR      3
IT      6
Name: Experience, dtype: int64
```

```
In [67]: c=df.agg({'Salary': ['sum']})
print(c)
```

```
Salary
sum  170000
```

```
In [112]: import pandas as pd
data = {
    'Student': ['John', 'Emma', 'Sam', 'Olivia'],
    'Marks': [80, None, 75, 90]
}
df=pd.DataFrame(data)
a=df.fillna(0)
print(a)
print("\n")
print(df.dropna())
```

```
Student Marks
0    John   80.0
1    Emma    0.0
2     Sam   75.0
3  Olivia   90.0
```

```
Student Marks
0    John   80.0
2     Sam   75.0
3  Olivia   90.0
```

```
In [91]: b=df.sort_values(by='Marks',ascending=False)
print(b)
```

	Student	Marks
3	Olivia	90.0
0	John	80.0
2	Sam	75.0
1	Emma	NaN

```
In [99]: c=df.to_csv('students.csv')
print(c)
```

None

```
In [100]: d=pd.read_csv('students.csv')
print(d)
```

	Unnamed: 0	Student	Marks
0	0	John	80.0
1	1	Emma	NaN
2	2	Sam	75.0
3	3	Olivia	90.0

```
In [101]: e=df.set_index('Student')
print(e)
```

	Marks
Student	
John	80.0
Emma	NaN
Sam	75.0
Olivia	90.0

```
In [102]: f=df.reset_index()
print(f)
```

	index	Student	Marks
0	0	John	80.0
1	1	Emma	NaN
2	2	Sam	75.0
3	3	Olivia	90.0

```
In [106]: import pandas as pd
data = {
    'Product': ['Laptop', 'Tablet', 'Smartphone', 'Monitor', 'Keyboard'],
    'Price': [70000, 30000, 25000, 15000, 2000],
    'Stock': [10, 25, 50, 15, 100]
}
df = pd.DataFrame(data)
print(df.loc[[0,2]])
```

	Product	Price	Stock
0	Laptop	70000	10
2	Smartphone	25000	50

```
In [113]: import pandas as pd
data = {
    'Subject': ['Math', 'Science', 'English'],
    'Marks': [88, 92, 85]
}
df=pd.DataFrame(data,index=['Student1', 'Student2', 'Student3'])
print(df.loc['Student2'])
```

```
Subject    Science
Marks              92
Name: Student2, dtype: object
```